EMMANUEL KONAN

Cybersecurity Engineer

@emmanuel.konan.contact | Portfolio | LinkedIn | GitHub

IT/IoT/OT cybersecurity engineer, I work on risk analysis, security audits, industrial systems and embedded security. My career has led me to work across several sectors, notably energy, finance and the public sector, on both technical and cyber risk management issues.

RELEVANT PROFESSIONAL EXPERIENCE

Ornisec

Oct 2024 - Present

Auditor & Cybersecurity Consultant - IT/OT/IoT

Cybersecurity consultant and auditor working across public/administrative organizations, financial institutions, energy companies and IT/software companies, covering risk management, technical audits, industrial cybersecurity and compliance.

- Lead EBIOS-RM risk analyses.
- Perform audits covering architecture, organization and governance.
- Conduct internal ISO/IEC 27001 audits.
- Support clients on regulatory compliance (GDPR, NIS, AI Act).
- Participate in the preparation and facilitation of cybersecurity crisis exercises.
- Design and deliver cybersecurity training to client companies, covering technical, governance and awareness topics for both expert and non-expert audiences.
- Lead the firm's AI adoption initiative, identifying use cases and supporting their secure integration into consulting activities.
- Act as cybersecurity referent for connected devices at Enedis.
- Define security requirements and conduct qualification and audits of industrial, embedded and connected equipment.
- Integrate post-quantum cryptography (PQC) algorithms into environments with limited compute and storage resources.

Enedis

Oct 2021 - Sep 2024

IT/OT Cybersecurity Engineer - Work-study Sep 2023 - Sep 2024

Within a multidisciplinary cybersecurity team, I worked on the security of embedded systems, smart metering and critical infrastructure.

- Assessed security risks and threat models related to security modules embedded in industrial systems.
- Analyzed defense-in-depth architectures, hardware security (CPU, memory, buses) and potential vulnerabilities.
- Conducted research on cryptographic mechanisms, SSH protocols (RFC 4250-4254), PLC and ISO/IEC 7816 smart-card standards.
- Performed vulnerability research and penetration-testing activities on embedded environments.
- Modified the OpenSSH client in C for custom security testing and information gathering.
- Developed Python and Bash scripts to automate security testing and analysis.

Non-Technical Losses Project Manager - Work-study Oct 2021 - Aug 2023

Within Enedis' B2B division, my work focused on reducing non-technical losses and fighting electricity fraud.

- Performed data mining and consumption analysis to identify potential fraud and abnormal electricity usage.
- Automated monthly reporting and KPIs using VBA, improving performance monitoring.
- Managed a national project introducing a new feature within the electricity distribution network.
- Conducted security and compliance assessments of internal business tools (DPIA).
- Supported the development of Enedis' B2B strategic plan and internal awareness initiatives.
- Completed a field immersion with operational technicians, gaining practical knowledge of electricity distribution, non-technical losses and the French energy market.

EDUCATION

CentraleSupélec

Engineering Degree - Computer Science & Cybersecurity 2021-2024

GPA 4.23/5.0. General engineering curriculum spanning computer science, cybersecurity, electronics, data/AI and management.

University of Rennes

Advanced Hardware Security - Master 2 coursework 2023-2024

Coursework in embedded-system attacks, including side channels, glitching and JTAG.

Saint Aubin La Salle

Preparatory Classes - PTSI/PT 2019-2021

Mathematics, physics and engineering sciences. Ranked first of the class, highest honors.

Lycée Classique d'Abidjan, Côte d'Ivoire

French Baccalaureate (Series C) - Mathematics & Physics/Chemistry 2016-2019

Graduated with honors. Ranked first of the class (10th - 12th grade).

PUBLICATIONS & STANDARDS CONTRIBUTIONS

1. **EAP-PSK-256 - Cryptographic protocol contribution** IETF Internet-Draft, 2026. Coauthor of an individual Internet-Draft targeting quantum-resistant authentication for constrained devices, using pre-shared keys, AES-256 and NIST SP 800-108 key derivation. (Work in progress). datatracker.ietf.org/doc/draft-eap-psk-256

SKILLS & INTERESTS

Certifications: ANSSI PASSI Qualified Auditor - organizational and physical audit (2026); PECB ISO 27001 Lead Auditor (2026); eJPT (2025)

Languages: French; English - TOEIC 910/990, Linguaskill C1

Core strengths: Leadership; Proactivity; Solution-oriented mindset; Adaptability; Time management & organization

Community:

- Digital Club Member, Elit Club (Jan 2025 - Present)
- Business Development Lead & IT Consultant, African Engineers Network (Dec 2024 - Present)
- Private Tutor, Superprof (Jan 2022 - Jun 2024)
- Treasurer & IT Support, Rézo Metz-Rennes Fédérés (Sep 2021 - Sep 2024)